

Product 2 — Scope Sheet

Product family • 5 members • Deliverable: A working web or mobile application.

For your family: Your Week 2 target is one complete user journey working end to end, not several half-built screens.

1. The community user

Who exactly is this for? Name the group, and where they are. "The community" is not specific enough.

Two groups, both in Ghana. Mentees: university students, recent graduates, bootcamp alumni, and career switchers trying to break into software engineering, data, product management, UX/design, or related tech roles. Mentors: working tech professionals in Ghana (software engineers, data practitioners, product managers, designers, engineering managers) willing to give short, structured guidance.

2. The problem, as it happens today

What do they currently do, and what goes wrong? Two or three sentences.

Right now, people trying to enter tech careers rely on cold outreach on LinkedIn, which gets very low response rates, or generic online advice that does not fit their specific background. University career services and informal networks often lack depth in current industry practice. As a result, capable people waste time, lose confidence, or make poorly informed decisions about skills and roles simply because they cannot get practical guidance from someone who has already walked the path.

3. What we are building

One sentence describing the product, then what changes for the user once it exists.

Pathfind is a web platform where mentees search a directory of verified tech mentors and send a structured mentorship request. Once it exists, a mentee no longer needs a personal connection or a lucky cold message to reach a real tech professional; they can find someone relevant and ask for specific help directly.

4. Core feature — working end to end by Friday 11 September (Gate 2)

The single most important thing your product does. Just one.

A searchable, filterable mentor directory where a mentee can find a mentor by role or expertise and send a structured mentorship request, end to end, from search to a request landing with the mentor.

5. Explicitly out of scope

What you have decided NOT to build. Teams that skip this always overrun.

- In-app messaging or chat (an accepted request hands off to email instead).
- Calendar or scheduling integration (mentors state general availability as free text, not a booking calendar).
- Password reset flows, email verification links, or OAuth/social login (plain email and password only for this build).
- A ratings and review system (a verified badge is used as the trust signal instead).

6. Who owns what

Every person needs something specific. Fill the right-hand column together.

Name	Track	Role	What they own
Mustapha Haadi	DevOps	Team Lead	Infrastructure: repo, VPC, ECR/ECS, CI/CD pipeline. Supports the team across blockers.
Edward Kamasah	Frontend	Builder	Mentor directory UI, search and filter, mentor profile pages.
Faith Ugwo Oghenetega	Frontend	Builder	Mentorship request flow (data model and API) and the mentee status dashboard.
Margaret Amanfu	DevOps	Foundations	RDS database setup, CI/CD support, monitoring, and email notifications (SES).
Stephen Nyarko Mensah	Fullstack	Foundations	Authentication endpoints (sign-up, sign-in, password hashing) and the request accept/decline logic.

7. Offline-first check

Question	Your answer
Does the product work with no or poor internet?	Core browsing works with poor connectivity. Previously loaded mentor profiles and directory listings remain viewable through basic caching. Search against fresh data, sign-up, and sending a request need a live connection.
Which parts need a connection, and what happens without one?	Sign-up, sign-in, live search, and sending or responding to a request need a connection. Without one, the app shows cached, previously loaded content and a clear "you are offline" notice rather than breaking.
Any heavy dependency (large downloads, paid API, cloud-only service)?	No. Everything runs on AWS free tier (S3, CloudFront, EC2/ECS, RDS). No paid third-party API is required.
If connectivity fails during the Week 5 demo, how will you still show it?	We will have a local build and a screen recording of the full flow ready as backup, so a venue connectivity issue cannot derail the presentation.

8. Tools and stack

What you will build it with. Everything must run on your own machines.

Frontend: React with TypeScript, built with Vite. Backend: Python (FastAPI), containerized with Docker. Database: PostgreSQL on AWS RDS, in a private subnet within our own VPC. Deployment: frontend on S3 and CloudFront; backend via ECR and ECS (EC2 launch type). Auth: custom email and password endpoints built in FastAPI. Email: AWS SES for request notifications. Secrets: AWS Secrets Manager. Monitoring: CloudWatch logs. CI/CD: GitHub Actions. All infrastructure is configured manually, no Terraform, and documented as we go. All AWS usage stays within the free tier.

9. Milestone plan

By when	What will be true
Fri 11 Sep — Gate 2	The core feature works end to end: a mentee can search the mentor directory and send a structured mentorship request, and it reaches the mentor.
Fri 18 Sep — Gate 3	Feature-complete and frozen: verification, email notifications, the mentee dashboard, and the verified badge are all working. No new features after this date.
Week 5 — what we demo	A live walkthrough: a mentee searching the directory, sending a request, a mentor accepting it, and both sides receiving the email handoff, backed by a real AWS deployment.

10. Risks and assumptions

What could stop you, and what you are assuming is true. Include anything outside your control.

Risks: venue or demo-day connectivity issues (mitigated by an offline fallback plan); real mentors may be slow to respond when invited near the end of the build, limiting authentic data in the final demo; team members balancing this alongside other commitments during the programme.

Assumptions: we can source a small number of real tech professionals willing to create mentor profiles before Week 4; all team members have reliable laptop and internet access for the programme; AWS free tier is sufficient for the scale of this demo.

Approval — Programme Lead use only

Decision: Approved / Approved with changes / Resubmit

Date: 04/09/26

Conditions or changes required:

Signed: 